Delta Sleep Inducing Peptide (DSIP) Available for Research Use Only
Most common 100 mcg intranasal, or 2.5 mg SC, Nightly Before Sleep

Some claim microdosing it is great for their sleep, others say it doesn't work at all. Early human trials were positive, subsequent ones showed no effect. Human trials used a very different dose than what's discussed in forums.

💉 💉 Probably safe (some human evidence)

0️⃣ Inconclusive (Evidence signals are mixed with some showing it doesn't work)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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Delta Sleep-Inducing Peptide (**DSIP**, sequence Trp-Ala-Gly-Gly-Asp-Ala-Ser-Gly-Glu) is a naturally occurring nonapeptide first identified for its association with **slow-wave (delta) sleep**. It has been investigated in **human clinical trials, animal models, and in vitro systems** for effects on sleep architecture, stress-axis regulation, analgesia, and neuroendocrine modulation. People who use DSIP subcutaneously generally seek **improved deep sleep, improve recovery, and modulate stress-related sleep disruption**.

[Delta sleep, sometimes called deep sleep *is the* ***deepest non-REM sleep stage*** *dominated by high-amplitude* ***delta waves****, marked by very low arousal threshold,* ***physical restoration*** *and* ***growth-hormone release.*** *In contrast,* ***REM sleep is characterized by rapid eye movements, vivid dreaming, and near-complete muscle atonia****, with high cortical activity important for* ***emotional processing and memory consolidation****.* ***Light sleep is a transitional, easily-arousable phase*** *that prepares the brain for deeper sleep and supports memory processing.]*

Evidence establishing the long-term safety of DSIP in humans is limited to a few available studies which are small, old, and do not include modern chronic-use toxicology. Widespread FRU use of DSIP provides only a **weak, indirect safety signal.**

Evidence to improving sleep is inconsistent at best. Early human studies reported **modest improvements in certain sleep parameters**, but later and better-controlled trials failed to reproduce these effects, leaving the overall evidence for DSIP as a sleep aid **weak and inconsistent**. Community use reports are also mixed.

Dose Evidence

DSIP is a small peptide subject to **rapid proteolytic degradation** in plasma, which shortens systemic half-life. Research used **slow IV infusion** to achieve sufficient central exposure. However, today, subcutaneous (SC) administration is used by individuals to provide a sustained peripheral depot and to **facilitate practical dosing**; however, peer-reviewed human trials using SC DSIP are limited, and the pharmacokinetic (PK) profile after SC dosing in humans is not uncharacterized in the published literature.

Graf, R (1981) as well as the landmark human summary by *Schneider-Helmert & Schoenenberger (1983)* describes human experiments using intravenous injections and explicitly **reports the experimental IV dose as 25 nmol/kg**. These paper emphasizes slow injection. 25 nmol/kg given was given as a slow intravenous injection, which corresponds to ≈21.2 µg/kg (≈0.0212 mg/kg), i.e., **≈1.5 mg for a 70-kg subject**.

Graf, R (1981) identifies **U-shaped sensitivity to dose and to administration timing** and reports a **narrow effective dose window centering on 1.5 mg** for DSIP and emphasize that **slow IV infusion** and **dosing close to 1 hour before the sleep period** were important to observe somnogenic effects. They report **both under- and over-dosing can reduce efficacy**, consistent with a U-shaped dose-response. *Confidence in this pattern is limited because the human data is sparse.*

However, **1.5 mg given slowly IV is likely to produce greater and faster systemic exposure than 1.5 mg given SC**, but the exact difference cannot be determined without measured bioavailability and PK parameters. If we assume DSIP's SC subcutaneous **bioavailability falls between 80% and 30%**, the typical range for peptides, matching a **1.5 mg IV** exposure would theoretically require **SC DSIP doses between 2 mg to 5 mg**, however, absorption rate, local degradation, formulation, injection site, and nonlinearity can change peak concentrations and effects, so **only measured PK data can confirm true equivalence**.

This dose was tested only for short bursts of up to four consecutive night and described as **"normalizing sleep architecture."** Although not explicitly discussed, it is plausible that this **high-dose, short-burst, is intended for only occasional use.** Unfortunately, **neither paper describe patterns or cycles in more specificity than "4 consecutive days,"** leaving contemporary researchers following their dosing guidelines to experiment with patterns using these bursts monthly, or more often, or less often, or, maybe, just as needed.

Dose, Routines and Patterns

Evidence-based Dose

Basing dose on *Graf, R (1981)* and *Schneider-Helmert (1983)*, a plausible human dose would be

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- **2 mg – 5.0 mg SC nightly for up to four consecutive nights.**

While this is most consistent with how DSIP was studied in human trials, it is hardly ever discussed by current independent FRU community researchers.

Common Community Low Dose

The very different approach to dosing DSIP, common in FRU communities, is

- **100 mcg – 0.5 mg nightly**, **intranasal**, (with 250 mcg being often discussed.)

This range of doses is usually discussed within the context of cycles, such as

- **Cycle 4 – 8 weeks on**, then **4 - 8 weeks off**, often combined with
- **Five days on, two days off per week**, within the 4–8-weeks-on blocks.

DSIP SC administration is plausibly best-timed tens of minutes to a few hours before sleep for acute somnogenic effect. Acute effects might begin within ~15–90 minutes, and SC administration could create a prolonged depot-like absorption with effects persisting hours.

Benefits

Benefits discussed below have been found or described in some settings with some cohorts and individual subjects. However, **some human trials have sometimes found no measurable sleep benefit**. The range of results, from devoted-fan to no-observable-effects might be **due to placebo effects** of non-blinded, non-random early trials, or **due to subject variability that is not understood**.

- **Improved slow-wave (deep) sleep architecture. (**Animal models consistently show modulation of sleep stages, while Human clinical data are inconsistent with some reporting increases in delta-wave sleep or improved sleep continuity.)
- **Modulation of the hypothalamic-pituitary-adrenal (HPA) axis and stress hormones.** (Limited human endocrine studies report changes in stress-axis markers consistent with reduced arousal.)
- **Subjective improvements in recovery, mood, and daytime function** are reported by individual subjects.

Indications

- **Chronic fragmented sleep** or low proportion of deep (delta) sleep.
- **Stress-related insomnia** or HPA-axis dysregulation with sleep disturbance.
- Adjunctive use in **recovery protocols** where enhanced slow-wave sleep is desired (e.g., athletic recovery).
- Situations where **non-benzodiazepine modulation of sleep architecture is preferred**.

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Contraindications

- **Pregnancy and breastfeeding** due to insufficient safety data.
- **Severe uncontrolled psychiatric illness** where sleep-modulating agents may alter risk. (
- **Concurrent use of sedative hypnotics, strong CNS depressants, or other investigational sleep agents** without medical supervision.
- **Severe hepatic or renal impairment** where peptide clearance is unknown due to lack of PK data.

Side Effects

- **Vivid dreams or altered dream content.**
- **Transient daytime somnolence or morning grogginess.**
- **Mild headache, dizziness, or nausea.**
- **Transient autonomic changes, e.g., heart-rate alterations (observed in animal studies.**)

Drivers of Diminished Response and Mitigation Strategies

For low dose DSIP: For **Receptor desensitization** the risk is low because delta sleep inducing peptide modulates sleep related neuroendocrine pathways rather than acting as a single receptor agonist. For **Neutralizing antibodies** the risk is low because DSIP is a small endogenous peptide with limited reported immunogenicity. For **downstream pathway adaptation** chronic modulation of sleep regulatory circuits can lead to reduced responsiveness. For **System Level Physiological Adaptation** circadian and sleep homeostasis can attenuate effect. **Overall likelihood of waning is low to moderate.**

Mitigation is **cycling On Off dosing.**

Use with Low Dose Epitalon

Western FRU **low dosing of Epitalon for sleep** is a grassroots practice **built from anecdote, experimentation, and community lore** rather than from any peer-reviewed research or mechanistic biology. **Practitioners talk about Epitalon as if it were a nightly sleep-aid,** something taken in **100 – 300 mcg doses**, often internasal, before bed to **smooth circadian rhythm**, **deepen sleep**, or **reduce nighttime awakenings**. **100 mcg is the most common dose.**

Practitioners frame it as a **gentle, ongoing "somnific" support** rather than a therapeutic cycle. The logic, not framed in mechanistic biology, is that small, steady doses **might nudge melatonin production, stabilize sleep architecture, or maintain pineal signaling without the intensity of the Russian ten-to-twenty-day high-dose courses.**

Some practitioners experiment with **combining low dose DSIP and low dose Epitalon**.

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Any use combining DSIP and Epitalon for sleep falls completely self-experimentation, with the only guidance for patterns and doses coming from other individual self-experimenters.

Biological Mechanism

DSIP's mechanism of action is **multifactorial and incompletely defined**, with convergent lines of evidence from **in vitro assays, animal neurophysiology, and limited human endocrine studies**. The summary below describes plausible mechanisms consistent with literature and experimental models.

Electrophysiological and sleep-recording studies in animals indicate that **DSIP influences neuronal excitability in sleep-regulatory circuits**, particularly those that generate **delta oscillations**. DSIP appears to **enhance inhibitory tone** in relevant networks, which is consistent with observed **increases in slow-wave sleep** in some models. Plausibly, this may involve **modulation of GABAergic signaling** and downstream suppression of arousal-promoting pathways.

Preclinical endocrine studies show that DSIP can **alter hypothalamic release of corticotropin-releasing factors** and modulate pituitary ACTH secretion, producing downstream changes in adrenal corticosteroid output. This **HPA-axis modulation** provides a plausible mechanism for DSIP's effects on stress-related sleep disruption and for reported reductions in stress-associated arousal.

In vitro neuronal cultures and animal injury models suggest DSIP can **reduce excitotoxic damage, modulate calcium homeostasis, and influence neurotrophic signaling pathways**. These effects may underlie observed **neuroprotective outcomes** in preclinical studies and support theoretical benefits for recovery and resilience.

Animal studies report **transient autonomic changes** (e.g., heart-rate variability shifts) after DSIP administration, suggesting peripheral autonomic modulation that could influence sleep consolidation and restorative physiology.

References

Schneider-Helmert D, Schoenenberger GA. Effects of DSIP in man. Neuropsychobiology. 1983;9(4):197–206. doi:10.1159/000117964. (Key controlled human sleep experiments (single and repeated slow IV injections) reporting somnogenic effects and the 4-night normalization finding.)

Monnier C, Schoenenberger GA. (Discovery paper) Isolation/characterization of delta sleep-inducing peptide. Experientia. 1977. (The foundational discovery, isolation, and demonstration of delta-wave induction in rabbit models.)

Schoenenberger GA, Monnier C. (Sequence/characterization) Eur J Biochem. 1978. (Peptide sequencing/biochemical characterization that established DSIP as a nonapeptide.)

Graf R, et al. (Early human/animal pharmacology) Eur J Pharmacol. 1981. (Early pharmacologic and sleep-architecture studies that helped define dose/route effects in animals and small human series.)

Nakamura S, et al. (HPA axis / stress models) Life Sci. 1988. (Representative preclinical work showing DSIP modulation of stress hormones and HPA responses anchoring sleep effects to an underlying mechanism.)